

The Problem

Government and institutional facilities generate large amounts of data on energy, water, waste, air quality, safety, traffic, and asset usage. However, this data is often fragmented across different sources, making it difficult to analyze and turn into timely decisions.

The Challenge

Administrators need a smart system that can:

-  Understand what is happening
-  Detect what looks abnormal
-  Predict what may happen next
-  Recommend what action to consider

for the MVP, taking an engineering college/campus as the primary example is a very good strategy. You do **not** need to build separate complete solutions for a college, hospital, municipality, industry, etc.

The smarter approach is:

Build deeply for ONE facility → design the architecture to be configurable → demonstrate how the same engine can expand to other facility types.

Proposed Solution

An AI-powered Facility Intelligence Dashboard that converts fragmented facility data into actionable insights for administrators.

Main Objective

The ultimate objective is:

Turn facility data into useful, actionable decision-support information for administrators.

Not just:

Data → Graph

but:

Data → Analysis → Problem Detection → Prediction → Recommendation → Decision

● MANDATORY / CORE FEATURES

1.Data ingestion

→ Accept open, synthetic, or sensor-style data.

2.Trend visualization

→ Show how facility conditions/data change over time.

3.Anomaly detection

→ Identify unusual or abnormal patterns.

4.Short-term forecasting

→ Predict upcoming changes.

5.Plain-language recommendations

→ Explain findings in a way administrators can understand.

6.AI-generated insights

→ Convert analyzed data into meaningful insights.

7.Action recommendations

→ Suggest practical actions administrators can consider.

● CORE REQUIREMENT 8 – Scenario Simulation

The PS says a **strong prototype should combine** visualization with: **“scenario simulation”**.

Simple meaning

Allow the administrator to ask:

"What if we change something?"

● CORE REQUIREMENT 9 – Configurable / Adaptable Facility

The PS says the solution should be adaptable to:

- Hospitals
- Colleges
- Company campuses
- Public-sector units
- Industrial estates
- Municipal facilities
- Large mixed-use institutional zones

● CORE REQUIREMENT 10 – Role-Based Views

The guidelines say solutions:

“should support role-based views for administrators, operations teams, and sustainability officers.”

For example:



Administrator

- Overall facility health
- Major risks
- Priority actions



Operations Team

- Maintenance
- Equipment
- Waste
- Water



Sustainability Officer

- Energy
- Water
- Waste
- Emissions

● Example Modules – NOT ALL MANDATORY

This part is very important.

The PS says:

“Example modules may include...”

Therefore, these are examples of areas you can choose, not a requirement to build all of them.

Examples listed by the PS:

Module	What it could do
 Air-quality monitoring	Monitor/analyze air quality
 Waste-bin overflow prediction	Predict when bins may overflow
 Energy optimization	Find inefficient energy usage
 Water-use alerts	Detect unusual water consumption
 Traffic/parking hotspots	Identify congested areas
 Equipment utilization	Analyze equipment usage
 Safety incident trends	Analyze safety patterns
 Sustainability scorecards	Summarize sustainability performance

So:

You DON'T need all 8.

For a hackathon, you could select:

 Energy +  Water +  Waste

and implement them well.